

Subscripts t and xx represent the first time derivative and second position derivative of the function f . The factor α in the equation is called the *diffusivity* which is defined by the system under investigation. The Schrödinger equation (in dimensionless units) using the same notation as in equation (29) becomes

$$i\Psi_t = -\frac{1}{2}\Psi_{xx}. \quad (30)$$

Except for the imaginary number i , the equation (30) is identical to (29). Therefore, it is mathematically correct to proceed to solve (30) with the complex extension of the same tools used in the numerical method for the Diffusion Equation. Numerical methods¹³ solve the *PDE* by transforming the integral problem into an algebraic one that is computationally accessible. The *Crank-Nicholson method* is the preferred numerical algorithm used to solve the Schrödinger equation as a diffusion type equation. It is an implicit algorithm valid through second order in both space and time coordinates, so it is very stable¹³.

To make that equation equivalent to the escape problem, it is necessary to employ the mathematical concepts of *TBCs*. The discretized domain of the solution is constructed using the following finite difference grids

$$x_i = (i - 1) \Delta x \quad (31)$$

$$t^n = n\Delta t \ (\Delta t \text{ constant}) \quad (32)$$

with $i = 0, 1, 2, \dots, imax$ and $n = 1, 2, \dots, nmax$. The domain $D(x, t)$ is from 0 to L on the $x - axis$ and the solution is marching in a positive time direction on the $y - axis$. We introduce the following notation in the context of the finite difference method grid point $(i, t) \rightarrow (x_i, t^n)$, function $f(x_i, t^n) \rightarrow f_i^n$, first time derivative $\frac{\partial f_i^n}{\partial t} \rightarrow f_t|_i^n$, and second space derivative $\frac{\partial^2 f_i^n}{\partial x^2} \rightarrow f_{xx}|_i^n$.

The *2nd Order Central Space* approximation of the second derivative is:

$$f_{xx}|_i^n = \frac{f_{i+1}^n - 2f_i^n + f_{i-1}^n}{\Delta x^2} \quad (33)$$

To obtain second order precision in the time, a key point of the method is to estimate derivatives at half integral time steps. The *2nd Order Central Time* approximation of the